Supporting Mathematica File for Reciprocal Preferences and Expectations in International Agreements

The number of countries is denoted by “n” instead of “N” here.

Proposition 2: Efforts

Non-signatory i’s best-response

In[1]:=
```
qni[b_, c_, n_, k_, α_, η_, qqs_] = (b (n - 1) + α (k qqs - (n - 1) η)) / (c (n - 1) - α (n - k - 1))
```

Out[1]=
$$\frac{b\,(-1+n)+\alpha\,\left(k\,qqs-(-1+n)\,\eta\right)}{c\,(-1+n)-(-1-k+n)\,\alpha}$$

Signatories’ objective function

In[2]:=
```
Us[b_, c_, n_, k_, α_, η_, qqs_, qqn_] =
  b (k qqs + (n - k) qqn) - (c / 2) qqs ^ 2 + α (((k - 1) qqs + (n - k) qqn) / (n - 1) - η) (qqs + 1 - η)
```

Out[2]=
$$-\frac{c\,qqs^2}{2}+b\,\left((-k+n)\,qqn+k\,qqs\right)+\alpha\,(1+qqs-\eta)\,\left(\frac{(-k+n)\,qqn+(-1+k)\,qqs}{-1+n}-\eta\right)$$

In[3]:=
```
Solve[D[Us[b, c, n, k, α, η, qqs, qni[b, c, n, k, α, η, qqs]], qqs] == 0, qqs]
```

Out[3]=
$$\left\{\left\{qqs\to\left(-\frac{b\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}-\frac{\alpha\,\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}-b\,\left(k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)+\alpha\,\eta+\frac{(-k+n)\,\alpha^2\,\eta}{c\,(-1+n)-(-1-k+n)\,\alpha}+\frac{\alpha\,\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)\,\eta}{-1+n}\right)\Big/\left(-c+\frac{(-1+k)\,\alpha}{-1+n}+\frac{k\,(-k+n)\,\alpha^2}{(-1+n)\,\left(c\,(-1+n)-(-1-k+n)\,\alpha\right)}+\frac{\alpha\,\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}\right)\right\}\right\}$$

Signatory effort - Proposition 2(1):

In[4]:= $\mathrm{qs}[b\_, c\_, n\_, k\_, \alpha\_, \eta\_] =$

$$\mathrm{FullSimplify}\left[\left(-\frac{b\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}-\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}-b\left(k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)+\alpha\,\eta+\frac{(-k+n)\,\alpha^2\,\eta}{c\,(-1+n)-(-1-k+n)\,\alpha}+\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)\eta}{-1+n}\right)\Big/\left(-c+\frac{(-1+k)\,\alpha}{-1+n}+\frac{k\,(-k+n)\,\alpha^2}{(-1+n)\,(c\,(-1+n)-(-1-k+n)\,\alpha)}+\frac{\alpha\left(-1+k+\frac{k\,(-k+n)\,\alpha}{c\,(-1+n)-(-1-k+n)\,\alpha}\right)}{-1+n}\right)\right]$$

Out[4]=

$$\frac{b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta))}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}$$

Non-signatory effort (Proposition 2(1)):

In[5]:= `qn[b_, c_, n_, k_, α_, η_] = FullSimplify[qni[b, c, n, k, α, η, qs[b, c, n, k, α, η]]]`

Out[5]=

$$\frac{b\,(-1+n)-(-1+n)\,\alpha\,\eta+\frac{k\,\alpha\,(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)))}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}}{c\,(-1+n)+(1+k-n)\,\alpha}$$

How do effort levels change in treaty size k?

In[6]:= `FullSimplify[D[qs[b, c, n, k, α, η], k]]`

Out[6]=

$$\frac{c\,(c\,(-1+n)+\alpha)\,(b\,c\,(-1+n)-b\,(-2+n)\,\alpha-\alpha\,(\alpha+c\,(-1+\eta)))}{\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2}$$

In[7]:= $\mathrm{FullSimplify}\left[\mathrm{Solve}\left[\left(b\,c\,(-1+n)-b\,(-2+n)\,\alpha-\alpha\,(\alpha+c\,(-1+\eta))\right)==0,\ c\right]\right]$

Out[7]=

$$\left\{\left\{c\to\frac{\alpha\,(b\,(-2+n)+\alpha)}{b\,(-1+n)+\alpha-\alpha\,\eta}\right\}\right\}$$

In[8]:= `FullSimplify[D[qn[b, c, n, k, α, η], k]]`

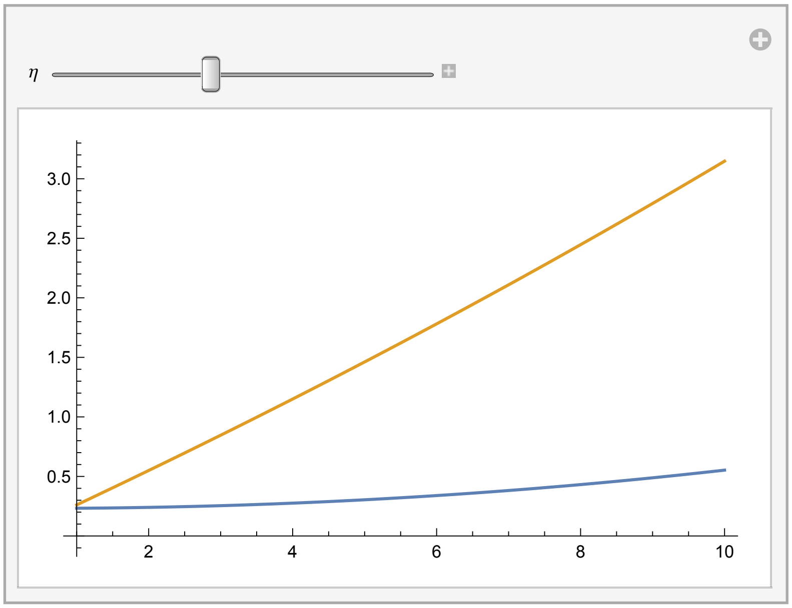
Out[8]=

$$\Big(\alpha\,\big(-c^3\,(-1+2\,k)\,(-1+n)^2+2\,c^2\,(2+k\,(-3+n)-n)\,(-1+n)\,\alpha+c\,(5-k\,(4+k)-6\,n+4\,k\,n+n^2)\,\alpha^2+2\,(-1+n)\,\alpha^3\big)\,(b\,(c-c\,n+(-2+n)\,\alpha)+\alpha\,(\alpha+c\,(-1+\eta)))\Big)\Big/\Big((c\,(-1+n)+(1+k-n)\,\alpha)^2\,\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2\Big)$$

These derivatives are difficult to see whether the efforts are increasing or decreasing in k. However, simple plots for any feasible parameter values show that both efforts are increasing in k; One example is shown below - **Proposition 2(3)**

In[9]:=
```
Manipulate[
 Plot[{qn[0.25, 1, 10, k, 0.1, η], qs[0.25, 1, 10, k, 0.1, η]}, {k, 1, 10}], {η, 0, 1}]
```

Out[9]=



The following difference is used in the proof of **proposition 2(4)**:

In[10]:=
```
FullSimplify[qs[b, c, n, k, α, η] - qn[b, c, n, k, α, η]]
```

Out[10]=
$$\frac{(-1+n)\,(c\,(-1+k)+\alpha)\,(b\,c\,(-1+n)-b\,(-2+n)\,\alpha-\alpha\,(\alpha+c\,(-1+\eta)))}{(c\,(-1+n)+(1+k-n)\,\alpha)\,(c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2)}$$

At what k, signatories exert full effort **(Proposition 2(5)**:

In[11]:=
```
FullSimplify[Solve[qs[b, c, n, k, α, η] == 1, k]]
```

Out[11]=
$$\left\{\left\{k\to\frac{c^2\,(-1+n)+c\,\alpha\,(4+n\,(-1+\eta)-2\,\eta)+\alpha\,(-b\,n+\alpha\,(-3+2\,\eta))}{c\,(b\,(-1+n)-\alpha\,(-2+\eta))}\right\}\right\}$$

Proposition 3

Finding partial derivatives of signatory and non-signatory abatements with respect to α:

In[12]:=
```
DsigAbateAlpha[b_, c_, n_, k_, α_, η_] = FullSimplify[D[qs[b, c, n, k, α, η], α]]
```

Out[12]=
$$\left(2\,b\,n\,\alpha^2+c\,\alpha\,(4\,b\,k\,(-1+n)+\alpha\,(1+k-n-2\,\eta))-c^3\,(-1+n)\,(1-k+(-2+k+n)\,\eta)+c^2\,(-1+n)\,(b\,(n+k\,(-3+k+n))+2\,\alpha-4\,\alpha\,\eta)\right)\Big/\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2$$

In[13]:= `DnonsigAbateAlpha[b_, c_, n_, k_, α_, η_] = FullSimplify[D[qn[b, c, n, k, α, η], α]]`

Out[13]=
$$\frac{1}{\left(c\,(-1+n)+(1+k-n)\,\alpha\right)^2}\Bigg(-(1+k-n)\left(b\,(-1+n)-(-1+n)\,\alpha\,\eta+\frac{k\,\alpha\,\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,\left(\alpha-2\,\alpha\,\eta+c\,\left(-1+k-(-2+k+n)\,\eta\right)\right)\right)}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}\right)+\left(c\,(-1+n)+(1+k-n)\,\alpha\right)\left(\eta-n\,\eta+\frac{k\,\alpha\,\left(c-c\,k-b\,n-2\,\alpha+c\,(-2+k+n)\,\eta+4\,\alpha\,\eta\right)}{-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2}+\frac{k\,\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,\left(\alpha-2\,\alpha\,\eta+c\,\left(-1+k-(-2+k+n)\,\eta\right)\right)\right)}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}+\frac{k\,\alpha\,\left(c\,(-3+k+n)+4\,\alpha\right)\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,\left(\alpha-2\,\alpha\,\eta+c\,\left(-1+k-(-2+k+n)\,\eta\right)\right)\right)}{\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2}\right)\Bigg)$$

Evaluating them at α=0 and then solving for η:

In[14]:= `FullSimplify[DsigAbateAlpha[b, c, n, k, 0, η]]`
`FullSimplify[DnonsigAbateAlpha[b, c, n, k, 0, η]]`

Out[14]=
$$\frac{b\,\left(n+k\,(-3+k+n)\right)+c\,\left(-1+k-(-2+k+n)\,\eta\right)}{c^2\,(-1+n)}$$

Out[15]=
$$\frac{\frac{b\,\left(-1+(-1+k)\,k+n\right)}{-1+n}-c\,\eta}{c^2}$$

In[16]:=
$$\frac{\frac{b\,\left(-1+(-1+k)\,k+n\right)}{-1+n}-c\,\eta}{c^2}$$

Out[16]=
$$\frac{\frac{b\,\left(-1+(-1+k)\,k+n\right)}{-1+n}-c\,\eta}{c^2}$$

In[17]:= `FullSimplify[Solve[` $\frac{b\,\left(n+k\,(-3+k+n)\right)+c\,\left(-1+k-(-2+k+n)\,\eta\right)}{c^2\,(-1+n)}$ `== 0, η] ]`

`FullSimplify[Solve[` $\frac{\frac{b\,\left(-1+(-1+k)\,k+n\right)}{-1+n}-c\,\eta}{c^2}$ `== 0, η]]`

Out[17]=
$$\left\{\left\{\eta\to\frac{c\,(-1+k)+b\,\left(n+k\,(-3+k+n)\right)}{c\,(-2+k+n)}\right\}\right\}$$

Out[18]=
$$\left\{\left\{\eta\to\frac{b\,\left(-1+(-1+k)\,k+n\right)}{c\,(-1+n)}\right\}\right\}$$

These give us T_H and T_L, respectively.

Indirect Welfare Functions

Signatory

In[19]:=
```
FullSimplify[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]]]
```

Out[19]=

$$\Big(b^2 \left(c^2 (-1+n)^2 \left((-2+k) k+2 n\right)+2 c (-1+n) \left(k^2+k (-3+n)-(-3+n) n\right) \alpha+\left(4 k (-1+n)+(4-3 n) n\right) \alpha^2\right)+2 b \alpha \Big(\left(c (-1+n)+\alpha\right) \left(c \left((-2+k) k+n\right)+(2 k-n) \alpha\right)+\Big(-c^2 (-1+n) \left((-2+k) k+n^2\right)+c \left(2 k-k^2 n+n \left(2+(-4+n) n\right)\right) \alpha+2 n (-1-k+n) \alpha^2\Big) \eta\Big)+\alpha \Big(\alpha^3+2 c^3 (-1+n)^2 (-1+\eta) \eta+2 c \alpha^2 \left(-1+k (-1+\eta)^2+\eta (-1+2 n+\eta-n \eta)\right)+c^2 \alpha \left(1-2 k (-1+\eta)^2+k^2 (-1+\eta)^2+\eta \left(4+2 (-4+n) n-\left(4+(-6+n) n\right) \eta\right)\right)\Big)\Big) \Big/ \Big(2 \left(c (-1+n)+(1+k-n) \alpha\right) \left(c^2 (-1+n)-c (-3+k+n) \alpha-2 \alpha^2\right)\Big)$$

Non-signatory

In[20]:=
```
Un[b_, c_, n_, k_, α_, η_, qqs_, qqn_] =
  b (k qqs + (n - k) qqn) - (c / 2) qqn^2 + α ((k qqs + (n - k - 1) qqn) / (n - 1) - η) (qqn + 1 - η)
```

Out[20]=

$$-\frac{c\, qqn^2}{2}+b \left((-k+n)\, qqn+k\, qqs\right)+\alpha\, (1+qqn-\eta) \left(\frac{(-1-k+n)\, qqn+k\, qqs}{-1+n}-\eta\right)$$

In[21]:=
```
FullSimplify[Un[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]]]
```

Out[21]=

$$\Big(\alpha \Big(b \left(c^2 (-1+n)^2+c \left(3+(-1+k) k-n\right) (-1+n) \alpha+\left(2+(-2+k) n\right) \alpha^2\right)-\left(c^2 (-1+n)^2-c (-3+n) (-1+n) \alpha+(2+k-2 n) \alpha^2\right) \left(\alpha+c (-1+\eta)\right)\Big) \Big(b \left(c^2 (-1+n) \left(-1+(-1+k) k+n\right)+c \left(-3-2 k+k^2+(4+k) n-n^2\right) \alpha+2 (1+k-n) \alpha^2\right)+c \Big(-c^2 (-1+n)^2 \eta+\alpha^2 \left(k-2 (1+k-n) \eta\right)+c \alpha \left((-1+k) k+\left(3+k-k^2-4 n+n^2\right) \eta\right)\Big)\Big)\Big) \Big/ \Big(\left(c (-1+n)+(1+k-n) \alpha\right)^2 \left(-c^2 (-1+n)+c (-3+k+n) \alpha+2 \alpha^2\right)^2\Big)-\frac{c \left(b (-1+n)-(-1+n) \alpha \eta+\frac{k \alpha \left(b c k (-1+n)+b n \alpha+\alpha \left(\alpha-2 \alpha \eta+c \left(-1+k-(-2+k+n) \eta\right)\right)\right)}{c^2 (-1+n)-c (-3+k+n) \alpha-2 \alpha^2}\right)^2}{2 \left(c (-1+n)+(1+k-n) \alpha\right)^2}+b \left(\frac{k \left(b c k (-1+n)+b n \alpha+\alpha \left(\alpha-2 \alpha \eta+c \left(-1+k-(-2+k+n) \eta\right)\right)\right)}{c^2 (-1+n)-c (-3+k+n) \alpha-2 \alpha^2}+\frac{(-k+n) \left(b (-1+n)-(-1+n) \alpha \eta+\frac{k \alpha \left(b c k (-1+n)+b n \alpha+\alpha \left(\alpha-2 \alpha \eta+c \left(-1+k-(-2+k+n) \eta\right)\right)\right)}{c^2 (-1+n)-c (-3+k+n) \alpha-2 \alpha^2}\right)}{c (-1+n)+(1+k-n) \alpha}\right)$$

Self-interested non-signatory and signatory:

In[22]:=
```
SelfUn[b_, c_, n_, k_] =
  FullSimplify[Un[b, c, n, k, 0, η, qs[b, c, n, k, 0, η], qn[b, c, n, k, 0, η]]]
SelfUs[b_, c_, n_, k_] = FullSimplify[
   Us[b, c, n, k, 0, η, qs[b, c, n, k, 0, η], qn[b, c, n, k, 0, η]]]
```

Out[22]=

$$\frac{b^2 \left(-1+2 (-1+k) k+2 n\right)}{2 c}$$

Out[23]=

$$\frac{b^2 \left((-2+k) k+2 n\right)}{2 c}$$

Stable coalition size for self-interested:

In[24]:=
```
FullSimplify[Solve[SelfUs[b, c, n, k] - SelfUn[b, c, n, k - 1] == 0, k]]
```
Out[24]= $\{\{k \to 1\}, \{k \to 3\}\}$

Lemma 1

The partial derivative of signatory indirect welfare with respect to k:

In[25]:=
```
FullSimplify[D[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]], k]]
```
Out[25]= $$\frac{(c\,(-1+k)+\alpha)\,(c\,(-1+n)+\alpha)^2\,\left(b\,\left(c-c\,n+(-2+n)\,\alpha\right)+\alpha\,(\alpha+c\,(-1+\eta))\right)^2}{\left(c\,(-1+n)+(1+k-n)\,\alpha\right)^2\,\left(-c^2\,(-1+n)+c\,(-3+k+n)\,\alpha+2\,\alpha^2\right)^2}$$

Coalition size minimizing signatory's indirect welfare:

In[26]:=
```
FullSimplify[
 Solve[D[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]], k] == 0, k]]
```
Out[26]= $\left\{\left\{k \to 1-\frac{\alpha}{c}\right\}\right\}$

are used in the **`proof of parts (1 and 2).`**

`At what k signatory and non-signatory indirect welfare are equal?`

In[27]:=
```
FullSimplify[Solve[Us[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]] -
    Un[b, c, n, k, α, η, qs[b, c, n, k, α, η], qn[b, c, n, k, α, η]] == 0, k]]
```
Out[27]= $$\left\{\left\{k \to 1-\frac{\alpha}{c}\right\},\left\{k \to -\frac{(-1+n)\,(c-\alpha)\,\left(c\,(-1+n)+2\,\alpha\right)}{c^2\,(-1+n)^2+2\,c\,(-1+n)\,\alpha+2\,\alpha^2}\right\}\right\}$$

The latter is negative. So, they are equal at $k = 1 - \frac{\alpha}{c}$. **This proves the part 4 and it is useful to prove the parts 3 and 5.**

Lemma 2

The partial derivative of signatory indirect utility wrt qn

In[73]:=
```
D[Us[b, c, n, k, α, η, qs, qn], qn]
```
Out[73]= $$b\,(-k+n)+\frac{(-k+n)\,\alpha\,(1+qs-\eta)}{-1+n}$$

The partial derivative of non-signatory effort wrt alpha

In[35]:=
```
D[qn[b, c, n, k, α, η], α]
```

Out[35]=

$$\frac{1}{c\,(-1+n)+(1+k-n)\,\alpha}\left(-(-1+n)\,\eta+\frac{k\,\alpha\,\left(b\,n+\alpha+\alpha\,(1-2\,\eta)-2\,\alpha\,\eta+c\,\left(-1+k-(-2+k+n)\,\eta\right)\right)}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}-\frac{k\,\left(-c\,(-3+k+n)-4\,\alpha\right)\,\alpha\,\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,\left(\alpha-2\,\alpha\,\eta+c\,\left(-1+k-(-2+k+n)\,\eta\right)\right)\right)}{\left(c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2\right)^2}+\frac{k\,\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,\left(\alpha-2\,\alpha\,\eta+c\,\left(-1+k-(-2+k+n)\,\eta\right)\right)\right)}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}\right)-\frac{(1+k-n)\,\left(b\,(-1+n)-(-1+n)\,\alpha\,\eta+\frac{k\,\alpha\,(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)))}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}\right)}{\left(c\,(-1+n)+(1+k-n)\,\alpha\right)^2}$$

The partial derivative of signatory indirect utility wrt qn (qs at optimal) times the partial derivative of non-signatory effort wrt alpha

In[43]:=
```
derUsqnalpha[b_, c_, n_, k_, α_, η_] :=
 (b (-k + n) + ((-k + n) α (1 + qs[b, c, n, k, α, η] - η))/(-1 + n))
  (((-(-1 + n) η + (k α (b n + α + α (1 - 2 η) - 2 α η + c (-1 + k - (-2 + k + n) η)))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2) -
      (k (-c (-3 + k + n) - 4 α) α (b c k (-1 + n) + b n α +
          α (α - 2 α η + c (-1 + k - (-2 + k + n) η)))) / (c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2)^2 +
      (k (b c k (-1 + n) + b n α + α (α - 2 α η + c (-1 + k - (-2 + k + n) η))))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2)) /
    (c (-1 + n) + (1 + k - n) α) -
   ((1 + k - n) (b (-1 + n) - (-1 + n) α η + (k α (b c k (-1 + n) + b n α + α (α - 2 α η + c (-1 + k - (-2 + k + n) η))))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2)))/(c (-1 + n) + (1 + k - n) α)^2)
```

Evaluating at alpha = 0.

In[44]:=
```
FullSimplify[derUsqnalpha[b, c, n, k, 0, η]]
```

Out[44]=

$$-\frac{b\,(k-n)\,\left(b\,\left(-1+(-1+k)\,k+n\right)-c\,(-1+n)\,\eta\right)}{c^2\,(-1+n)}$$

The partial derivative of signatory indirect utility wrt qs

In[42]:=
```
D[Us[b, c, n, k, α, η, qs, qn], qs]
```

Out[42]=

$$b\,k-c\,qs+\frac{(-1+k)\,\alpha\,(1+qs-\eta)}{-1+n}+\alpha\left(\frac{(-k+n)\,qn+(-1+k)\,qs}{-1+n}-\eta\right)$$

The partial derivative of signatory effort wrt alpha

In[45]:=
```
D[qs[b, c, n, k, α, η], α]
```

Out[45]=
$$\frac{b\,n+\alpha+\alpha\,(1-2\,\eta)-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}-\frac{(-c\,(-3+k+n)-4\,\alpha)\,(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)))}{(c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2)^2}$$

The partial derivative of signatory indirect utility wrt qs (qs and qn at optimal) times t partial derivative of signatory effort wrt alpha

In[50]:=
$$\text{derUsqsalpha}[b\_, c\_, n\_, k\_, \alpha\_, \eta\_] := \left(b\,k-c\,\text{qs}[b,c,n,k,\alpha,\eta]+\frac{(-1+k)\,\alpha\,(1+\text{qs}[b,c,n,k,\alpha,\eta]-\eta)}{-1+n}+\alpha\left(\frac{(-k+n)\,\text{qn}[b,c,n,k,\alpha,\eta]+(-1+k)\,\text{qs}[b,c,n,k,\alpha,\eta]}{-1+n}-\eta\right)\right)\left(\frac{b\,n+\alpha+\alpha\,(1-2\,\eta)-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)}{c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2}-\frac{(-c\,(-3+k+n)-4\,\alpha)\,(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\,(\alpha-2\,\alpha\,\eta+c\,(-1+k-(-2+k+n)\,\eta)))}{(c^2\,(-1+n)-c\,(-3+k+n)\,\alpha-2\,\alpha^2)^2}\right)$$

Evaluating at alpha = 0.

In[47]:=
```
FullSimplify[derUsqsalpha[b, c, n, k, 0, η]]
```

Out[47]=
$$\frac{(b\,k-c\,\text{qs})\,(b\,(n+k\,(-3+k+n))+c\,(-1+k-(-2+k+n)\,\eta))}{c^2\,(-1+n)}$$

Combining both:

In[56]:=
```
derUsalphazero[b_, c_, n_, k_, η_] =
 FullSimplify[derUsqnalpha[b, c, n, k, 0, η] + derUsqsalpha[b, c, n, k, 0, η]]
```

Out[56]=
$$-\frac{b\,(k-n)\,(b\,(-1+(-1+k)\,k+n)-c\,(-1+n)\,\eta)}{c^2\,(-1+n)}$$

Finding the critical level of eta that equate the term above to zero:

In[55]:=
```
Solve[derUsalphazero[b, c, n, k, η] == 0, η]
```

Out[55]=
$$\left\{\left\{\eta\to\frac{b\,(-1-k+k^2+n)}{c\,(-1+n)}\right\}\right\}$$

which completes the **`part 1`**.

The partial derivative of non-signatory indirect utility wrt qs

In[59]:=
```
D[Un[b, c, n, k, α, η, qs, qn], qs]
```

Out[59]=
$$b\,k+\frac{k\,\alpha\,(1+\text{qn}-\eta)}{-1+n}$$

The partial derivative of signatory effort wrt alpha

In[60]:= `D[qs[b, c, n, k, α, η], α]`

Out[60]=
$$\frac{b\,n+\alpha+\alpha\left(1-2\,\eta\right)-2\,\alpha\,\eta+c\left(-1+k-\left(-2+k+n\right)\eta\right)}{c^2\,(-1+n)-c\left(-3+k+n\right)\alpha-2\,\alpha^2}-\frac{\left(-c\left(-3+k+n\right)-4\,\alpha\right)\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\left(\alpha-2\,\alpha\,\eta+c\left(-1+k-\left(-2+k+n\right)\eta\right)\right)\right)}{\left(c^2\,(-1+n)-c\left(-3+k+n\right)\alpha-2\,\alpha^2\right)^2}$$

The partial derivative of non-signatory indirect utility wrt qs (qn at optimal) times the partial derivative of signatory effort wrt alpha

In[63]:=
```
derUnqsalpha[b_, c_, n_, k_, α_, η_] :=
 FullSimplify[(b k + (k α (1 + qn[b, c, n, k, α, η] - η))/(-1 + n))
   ((b n + α + α (1 - 2 η) - 2 α η + c (-1 + k - (-2 + k + n) η))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2) -
    ((-c (-3 + k + n) - 4 α) (b c k (-1 + n) + b n α + α (α - 2 α η + c (-1 + k - (-2 + k + n) η))))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2)^2)]
```

Evaluating at alpha = 0.

In[64]:= `FullSimplify[derUnqsalpha[b, c, n, k, 0, η]]`

Out[64]=
$$\frac{b\,k\left(b\left(n+k\left(-3+k+n\right)\right)+c\left(-1+k-\left(-2+k+n\right)\eta\right)\right)}{c^2\,(-1+n)}$$

The partial derivative of non-signatory indirect utility wrt qn

In[65]:= `D[Un[b, c, n, k, α, η, qs, qn], qn]`

Out[65]=
$$b\left(-k+n\right)-c\,qn+\frac{\left(-1-k+n\right)\alpha\,(1+qn-\eta)}{-1+n}+\alpha\left(\frac{\left(-1-k+n\right)qn+k\,qs}{-1+n}-\eta\right)$$

The partial derivative of non-signatory effort wrt alpha

In[66]:= `D[qn[b, c, n, k, α, η], α]`

Out[66]=
$$\frac{1}{c\,(-1+n)+\left(1+k-n\right)\alpha}\left(-(-1+n)\,\eta+\frac{k\,\alpha\left(b\,n+\alpha+\alpha\left(1-2\,\eta\right)-2\,\alpha\,\eta+c\left(-1+k-\left(-2+k+n\right)\eta\right)\right)}{c^2\,(-1+n)-c\left(-3+k+n\right)\alpha-2\,\alpha^2}-\frac{k\left(-c\left(-3+k+n\right)-4\,\alpha\right)\alpha\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\left(\alpha-2\,\alpha\,\eta+c\left(-1+k-\left(-2+k+n\right)\eta\right)\right)\right)}{\left(c^2\,(-1+n)-c\left(-3+k+n\right)\alpha-2\,\alpha^2\right)^2}+\frac{k\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\left(\alpha-2\,\alpha\,\eta+c\left(-1+k-\left(-2+k+n\right)\eta\right)\right)\right)}{c^2\,(-1+n)-c\left(-3+k+n\right)\alpha-2\,\alpha^2}\right)-\frac{\left(1+k-n\right)\left(b\,(-1+n)-(-1+n)\,\alpha\,\eta+\frac{k\,\alpha\left(b\,c\,k\,(-1+n)+b\,n\,\alpha+\alpha\left(\alpha-2\,\alpha\,\eta+c\left(-1+k-\left(-2+k+n\right)\eta\right)\right)\right)}{c^2\,(-1+n)-c\left(-3+k+n\right)\alpha-2\,\alpha^2}\right)}{\left(c\,(-1+n)+\left(1+k-n\right)\alpha\right)^2}$$

The partial derivative of non-signatory indirect utility wrt qn times the partial derivative of non-signatory effort wrt alpha

In[68]:=
```
derUnqnalpha[b_, c_, n_, k_, α_, η_] := FullSimplify[
  (b (-k + n) - c qn[b, c, n, k, α, η] + ((-1 - k + n) α (1 + qn[b, c, n, k, α, η] - η))/(-1 + n) +
     α (((-1 - k + n) qn[b, c, n, k, α, η] + k qs[b, c, n, k, α, η])/(-1 + n) - η))
  ((-(-1 + n) η + (k α (b n + α + α (1 - 2 η) - 2 α η + c (-1 + k - (-2 + k + n) η)))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2) -
       (k (-c (-3 + k + n) - 4 α) α (b c k (-1 + n) + b n α + α (α - 2 α η +
             c (-1 + k - (-2 + k + n) η)))) / (c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2)^2 +
       (k (b c k (-1 + n) + b n α + α (α - 2 α η + c (-1 + k - (-2 + k + n) η))))/(c^2 (-1 + n) - c (-3 + k + n) α - 2 α^2)) /
     (c (-1 + n) + (1 + k - n) α) -
    ((1 + k - n) (b (-1 + n) - (-1 + n) α η + (k α (b c k (-1+n) + b n α + α (α-2 α η + c (-1+k-(-2+k+n) η))))/(c^2 (-1+n) - c (-3+k+n) α - 2 α^2)))/
     (c (-1 + n) + (1 + k - n) α)^2)]
```

Evaluating at alpha = 0.

In[69]:=
```
FullSimplify[derUnqnalpha[b, c, n, k, 0, η]]
```

Out[69]=
$$-\frac{b\,(1+k-n)\,\left(b\,(-1+(-1+k)\,k+n)-c\,(-1+n)\,\eta\right)}{c^2\,(-1+n)}$$

Combining both:

In[70]:=
```
derUnalphazero[b_, c_, n_, k_, η_] =
  FullSimplify[derUnqsalpha[b, c, n, k, 0, η] + derUnqnalpha[b, c, n, k, 0, η]]
```

Out[70]=
$$-\frac{2\,b\,(1+k-n)\,\left(b\,(-1+(-1+k)\,k+n)-c\,(-1+n)\,\eta\right)}{c^2\,(-1+n)}$$

In[71]:=
```
Solve[derUnalphazero[b, c, n, k, η] == 0, η]
```

Out[71]=
$$\left\{\left\{\eta\to\frac{b\,\left(-1-k+k^2+n\right)}{c\,(-1+n)}\right\}\right\}$$

The partial derivative of non-signatory indirect welfare wrt alpha at k-1 and evaluating at alpha = 0:

In[77]:=
```
derUnalphazero[b, c, n, k - 1, η]
```

Out[77]=
$$-\frac{2\,b\,(k-n)\,\left(b\,(-1+(-2+k)\,(-1+k)+n)-c\,(-1+n)\,\eta\right)}{c^2\,(-1+n)}$$

In[78]:=
```
Solve[derUnalphazero[b, c, n, k - 1, η] == 0, η]
```

Out[78]=
$$\left\{\left\{\eta\to\frac{b\,\left(1-3\,k+k^2+n\right)}{c\,(-1+n)}\right\}\right\}$$

Proposition 4

In[218]:=
```
derUsalphazero[b, c, n, 3, η]
```

Out[218]= $-\frac{b\,(3-n)\,(b\,(5+n)-c\,(-1+n)\,\eta)}{c^2\,(-1+n)}$

In[219]:=
```
derUnalphazero[b, c, n, 2, η]
```

Out[219]= $-\frac{2\,b\,(3-n)\,(b\,(1+n)-c\,(-1+n)\,\eta)}{c^2\,(-1+n)}$

Stability of Grand Coalition with Interior Solution

We showed in Proposition 2(5) that we need two conditions satisfied for an interior solution so that qi's are in (0,1).

1. $\alpha < \min\{b/\eta; (c-b)/(1-\eta)\}$
2. k < kbar

Furthermore, we need to make sure that the parameters we choose satisfy:

1. $c > b > 0$
2. $c > 2\alpha$
3. η in (0,1]

In[85]:=
```
kbar[b_, c_, n_, α_, η_] :=
```
$$\text{FullSimplify}\left[\frac{c^2\,(-1+n)+c\,\alpha\,(4+n\,(-1+\eta)-2\,\eta)+\alpha\,(-b\,n+\alpha\,(-3+2\,\eta))}{c\,(b\,(-1+n)-\alpha\,(-2+\eta))}\right]$$

Analyzing the condition U_s(N) ≥ U_n(N-1):

In[86]:=
```
FullSimplify[Us[b, c, n, n, α, η, qs[b, c, n, n, α, η], qn[b, c, n, n, α, η]] -
  Un[b, c, n, n - 1, α, η, qs[b, c, n, n - 1, α, η], qn[b, c, n, n - 1, α, η]]]
```

Out[86]= $-\left(\left(\left(c^3\,(-3+n)\,(-1+n)-2\,c^2\,(-3+n)\,(-2+n)\,\alpha+c\,(9-4\,n)\,\alpha^2-2\,\alpha^3\right)\left(b\,(c-c\,n+(-2+n)\,\alpha)+\alpha\,(\alpha+c\,(-1+\eta))\right)^2\right)\Big/\left(2\,c\,(c-2\,\alpha)\,\left(c^2\,(-1+n)-2\,c\,(-2+n)\,\alpha-2\,\alpha^2\right)^2\right)\right)$

Notice that there are two terms multiplied in the numerator. The second one is positive given that it is the squared. The denominator is positive since c>0, c>2α, and the squared term.
Thus, for this condition to hold, we need the first term in the numerator to be negative given that there is a minus in front of the whole term.

In[87]:=
```
term1[c_, n_, α_] := c³ (-3 + n) (-1 + n) - 2 c² (-3 + n) (-2 + n) α + c (9 - 4 n) α² - 2 α³
```

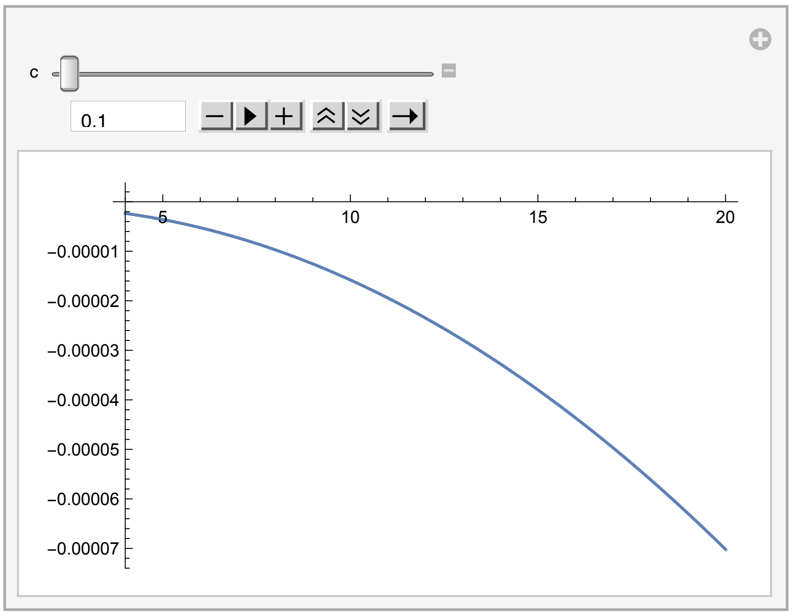
There are 4 terms in term1.
The first: it is always positive for n > 3; (zero for n = 1 or n = 3; negative for n = 2.) c amplifies the magnitude of this term.

The second: it is negative for n > 3. c and α amplify the magnitude of this term.
The third: it is negative for n ≥ 3. c and α amplify the magnitude of this term.
The fourth: it is always negative. α amplifies the magnitude of this term.
Thus, alpha always decreases term1 for n > 3. So, we should choose high alpha’s.

Strategy: search for parameter values such that term1 is negative as desired. Then, check whether the parameter values that satisfy term1<0 also satisfy interior solution.

In[88]:=
```
Manipulate[Plot[term1[c, n, 0.06] * 1 / 1000, {n, 4, 20}], {c, 0.1, 10}]
```

Out[88]=



By manipulating the parameter value of “c”, we see that for sufficiently small c term1 yields negative values.
n amplifies the impact.
A higher alpha expands the domain for c such that term1 yields negative values.
Some examples obtained using the manipulation and depicting graph for different alpha values:
alpha = 0.3, for all c < 0.6, term1 < 0; For c around 0.6+, term1 gets negative values for small n but positive values for large n.
alpha = 0.5, for all c < 1, term1 < 0; Similar argument above applies here and for the conditions below.
alpha = 0.6, for all c < 1.2, term1 < 0
alpha = 0.7, for all c < 1.4, term1 < 0
alpha = 0.8, for all c < 1.6, term1 < 0 though the impact of n is non-monotonic.
What about the smallest possible alpha?
Allowing for two-digits after comma, the smallest alpha that still produces term1 < 0 is 0.06. For all c < 0.12, term1 < 0.

For the existence of stable grand coalition with interior optimal efforts, we need to provide one example.
Take b=0.0309, c = 1.01, n = 10, alpha = 0.5, and eta = 0.8 .

The term1 and hence the condition U_s(N) ≥ U_n(N-1) holds:

In[89]:=
```
term1[1.01, 10, 0.5]
```
Out[89]=
```
-0.294137
```

And we have interior solutions at the grand coalition:

In[91]:=
```
FullSimplify[qs[.0309, 1.01, 10, 10, 0.5, 0.8]]
```
Out[91]=
```
0.9
```

In[92]:=
```
FullSimplify[qn[.0309, 1.01, 10, 10, 0.5, 0.8]]
```
Out[92]=
```
0.122847
```

But we need to make sure that effort levels are non-negative at k = N-1:

In[93]:=
```
FullSimplify[qs[.0307, 1.01, 10, 9, 0.5, 0.8]]
```
Out[93]=
```
-0.520774
```

In[94]:=
```
FullSimplify[qn[.0307, 1.01, 10, 9, 0.5, 0.8]]
```
Out[94]=
```
-0.623452
```

This shows that at these parameters, both qs and qn are considered negative when evaluating Un(N-1). So, making sure that the lowest value they can take is zero yields:

In[95]:=
```
FullSimplify[Us[.0307, 1.01, 10, 10, 0.5, 0.8, qs[.0307, 1.01, 10, 10, 0.5, 0.8],
   qn[.0307, 1.01, 10, 10, 0.5, 0.8]] - Un[.0307, 1.01, 10, 9, 0.5, 0.8, 0, 0]]
```
Out[95]=
```
0.00245
```

Therefore, the condition U_s(N) ≥ U_n(N-1) continues to hold when we make sure that effort levels are in [0,1) and found that there is a stable grand coalition with the optimal effort levels in the interior of their domains.

Strategic Use of Eta

In the paper, we assume N-1 countries set their truthful expectations “eta” while one country is strategically setting its expectations. Thus, we have a case of heterogenous expectations. In this supporting file, we first solve the game with possibly two types of countries different in their expectations, which we call them low and high expectations here. Then, assume that there is only one country of one of the types.

Let the number of countries be N, low types by l and, thus, the high types be N-l. Among k signatories, k_L and k_H be the number of low and high types, respectively.
The utility under symmetry, the equation (7): U(b, c, α, η, N, q_i, Q_-i) = b(Q_-i + qi) - (c/2)(qi)^2 +α(Q_-i/(N -1) - η)(q_i +1 - η)
Let’s solve the non-cooperative game.

The utility function of a country with LOW EXPECTATIONS is:

```
In[96]:= utilitynoncooplowetai[b_, c_, α_, ηL_, ηH_, n_, l_, qLn_, qHn_, qLi_] :=
  b ((n - l) qHn + (l - 1) qLn + qLi) - (c / 2) qLi ^ 2 +
   α (((n - l) qHn + (l - 1) qLn) / (n - 1) - ηL) (qLi + 1 - ηH)
```

The best-response function of country i with LOW EXPECTATION:

```
In[97]:= FullSimplify[
  Solve[D[utilitynoncooplowetai[b, c, α, ηL, ηH, n, l, qLn, qHn, qLi], qLi] == 0, qLi]]
```

$$\text{Out[97]=}\ \left\{\left\{\text{qLi} \to \frac{b + \frac{\alpha\,(n\,\text{qHn} - \text{qLn} + l\,(-\text{qHn} + \text{qLn}) + \eta L - n\,\eta L)}{-1+n}}{c}\right\}\right\}$$

In equilibrium, all identical types exert the same effort level:

```
In[98]:= FullSimplify[
  Solve[qLn == (b + (α (n qHn - qLn + l (-qHn + qLn) + ηL - n ηL)) / (-1 + n)) / c, qLn]]
```

$$\text{Out[98]=}\ \left\{\left\{\text{qLn} \to \frac{b\,(-1+n) + \alpha\,(-l\,\text{qHn} + n\,\text{qHn} + \eta L - n\,\eta L)}{c\,(-1+n) + \alpha - l\,\alpha}\right\}\right\}$$

The utility function of a country with HIGH EXPECTATIONS is:

```
In[99]:= utilitynoncoophighetai[b_, c_, α_, ηL_, ηH_, n_, l_, qLn_, qHn_, qHi_] :=
  b ((n - l - 1) qHn + l qLn + qHi) - (c / 2) qHi ^ 2 +
   α (((n - l - 1) qHn + l qLn) / (n - 1) - ηH) (qHi + 1 - ((n - 2) / (n - 1)) ηH - ηL / (n - 1))
```

The best-response function of country i with HIGH EXPECTATIONS:

```
In[100]:= FullSimplify[Solve[
   D[utilitynoncoophighetai[b, c, α, ηL, ηH, n, l, qLn, qHn, qHi], qHi] == 0, qHi]]
```

$$\text{Out[100]=}\ \left\{\left\{\text{qHi} \to \frac{b + \frac{\alpha\,((-1-l+n)\,\text{qHn} + l\,\text{qLn} + \eta H - n\,\eta H)}{-1+n}}{c}\right\}\right\}$$

In equilibrium, all identical types exert the same effort level:

```
In[101]:= FullSimplify[
  Solve[qHn == (b + (α ((-1 - l + n) qHn + l qLn + ηH - n ηH)) / (-1 + n)) / c, qHn]]
```

$$\text{Out[101]=}\ \left\{\left\{\text{qHn} \to \frac{b\,(-1+n) + \alpha\,(l\,\text{qLn} + \eta H - n\,\eta H)}{c\,(-1+n) + (1+l-n)\,\alpha}\right\}\right\}$$

Substituting one into the other yields the non-cooperative efforts with HIGH and LOW EXPECTATIONS:

$$\text{In[102]:=}\ \text{FullSimplify}\left[\text{Solve}\left[\text{qHn} == \frac{b\,(-1+n) + \alpha\left(l\left(\frac{b\,(-1+n) + \alpha\,(-l\,\text{qHn} + n\,\text{qHn} + \eta L - n\,\eta L)}{c\,(-1+n) + \alpha - l\,\alpha}\right) + \eta H - n\,\eta H\right)}{c\,(-1+n) + (1+l-n)\,\alpha}, \text{qHn}\right]\right]$$

$$\text{Out[102]=}\ \left\{\left\{\text{qHn} \to \frac{b\,(c\,(-1+n) + \alpha) + \alpha\,((c - c\,n + (-1+l)\,\alpha)\,\eta H - l\,\alpha\,\eta L)}{(c - \alpha)\,(c\,(-1+n) + \alpha)}\right\}\right\}$$

In[103]:= **FullSimplify[Solve[qLn == $\dfrac{b\,(-1+n)+\alpha\left((n-l)\left(\frac{b\,(-1+n)+\alpha\,(l\,qLn+\eta H-n\,\eta H)}{c\,(-1+n)+(1+l-n)\,\alpha}\right)+\eta L-n\,\eta L\right)}{c\,(-1+n)+\alpha-l\,\alpha}$, qLn]]**

Out[103]= $\left\{\left\{qLn\to\dfrac{b\,(c\,(-1+n)+\alpha)+\alpha\left(-n\,\alpha\,\eta H+l\,\alpha\,(\eta H-\eta L)+(-1+n)\,(-c+\alpha)\,\eta L\right)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}\right\}\right\}$

In[105]:= **noncoopeffhigh[b_, c_, α_, ηH_, ηL_, n_, l_] :=**
$$\frac{b\,(c\,(-1+n)+\alpha)+\alpha\left((c-c\,n+(-1+l)\,\alpha)\,\eta H-l\,\alpha\,\eta L\right)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$$

In[106]:= **noncoopefflow[b_, c_, α_, ηH_, ηL_, n_, l_] :=**
$$\frac{b\,(c\,(-1+n)+\alpha)+\alpha\left(-n\,\alpha\,\eta H+l\,\alpha\,(\eta H-\eta L)+(-1+n)\,(-c+\alpha)\,\eta L\right)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$$

Let the number of “low” types be 1, then the effort levels become as follows (**Equations 16 and 17 in the paper**):

In[107]:= **noncoopeffhigh[b, c, α, ηH, ηL, n, 1]**

Out[107]= $\dfrac{b\,(c\,(-1+n)+\alpha)+\alpha\left((c-c\,n)\,\eta H-\alpha\,\eta L\right)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$

In[108]:= **noncoopefflow[b, c, α, ηH, ηL, n, 1]**

Out[108]= $\dfrac{b\,(c\,(-1+n)+\alpha)+\alpha\left(-n\,\alpha\,\eta H+\alpha\,(\eta H-\eta L)+(-1+n)\,(-c+\alpha)\,\eta L\right)}{(c-\alpha)\,(c\,(-1+n)+\alpha)}$

From the best-responses qLn and qHn, one can see that qLn > qHn since *η*H > *η*L. Let us give some parameter values to understand how etas affect effort levels.

In[109]:= **FullSimplify[noncoopeffhigh[1, 5, 0.1, ηH, ηL, 10, 1]]**

Out[109]= $0.204082-0.0203629\,\eta H-0.0000452509\,\eta L$

In[110]:= **FullSimplify[noncoopefflow[1, 5, 0.1, ηH, ηL, 10, 1]]**

Out[110]= $0.204082-0.000407258\,\eta H-0.0200009\,\eta L$

Self-fulfilling Prophecy

Let us find the indirect utility levels. First, a country strategically sets its expectation under self-fulfilling prophecy assumption:

In[117]:=
```
indutilnoncooploweta[b_, c_, α_, ηH_, ηL_, n_, l_] :=
 FullSimplify[b ((n - l) ((b (c (-1 + n) + α) + α ((c - c n + (-1 + l) α) ηH - l α ηL))/((c - α) (c (-1 + n) + α))) +
     l ((b (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL))/((c - α) (c (-1 + n) + α)))) -
   (c/2) ((b (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL))/((c - α) (c (-1 + n) + α)))^2 +
   α (((n - l) ((b (c (-1 + n) + α) + α ((c - c n + (-1 + l) α) ηH - l α ηL))/((c - α) (c (-1 + n) + α))) + (l - 1)
        ((b (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL))/((c - α) (c (-1 + n) + α))))/(n - 1) -
     ηL) (((b (c (-1 + n) + α) + α (-n α ηH + l α (ηH - ηL) + (-1 + n) (-c + α) ηL))/((c - α) (c (-1 + n) + α))) + 1 - ηH)]
```

Country S' optimal strategic eta:

In[118]:=
```
FullSimplify[Solve[D[indutilnoncooploweta[b, c, α, ηH, ηL, n, 1], ηL] == 0, ηL]]
```

Out[118]=
```
{{ηL → (b (c (-1 + n) + α) (c^2 (-1 + n) + c α - (-1 + n) α^2) -
     c (c (-1 + n) - (-2 + n) α) (c^2 (-1 + n) (-1 + ηH) - c (-2 + n) α (-1 + ηH) +
        α^2 (1 + (-2 + n) ηH))) / (c α (c - c n + (-2 + n) α)^2)}}
```

The first two equations below show the optimal strategic eta "ηL" for some specific parameter values and in particular how it changes with the truthful expectations "ηH". The last two equations show the effort levels of the strategic and non-strategic countries. We evaluate these equations for a wide range of parameters and shared our findings on **equation (18), result 2 and the paragraph between the two.**

In[171]:=
```
FullSimplify[Solve[D[indutilnoncooploweta[.1, .3, 0.1, ηH, ηL, 10, 1], ηL] == 0, ηL]]
```

Out[171]=
```
{{ηL → 4.88643 - 3.42105 ηH}}
```

In[170]:=
```
FullSimplify[
 Solve[D[indutilnoncooploweta[0.1, 0.9, 0.2, ηH, ηL, 10, 1], ηL] == 0, ηL]]
```

Out[170]=
```
{{ηL → 5.24521 - 4.74615 ηH}}
```

In[115]:=
```
FullSimplify[noncoopeffhigh[1, 9, 0.1, ηH, ηL, 10, 1]]
```

Out[115]=
```
0.11236 - 0.0112221 ηH - 0.0000138544 ηL
```

In[116]:=
```
FullSimplify[noncoopefflow[1, 8, 0.1, ηH, ηL, 10, 1]]
```

Out[116]=
```
0.126582 - 0.000158008 ηH - 0.0125002 ηL
```

Self-Awareness

Strategic country's effort: $\frac{b\ (-1+n)+\alpha\ (-l\ qHn+n\ qHn+\eta-n\ \eta)}{c\ (-1+n)+\alpha-l\ \alpha}$

What others believe: $\frac{b\ (-1+n)+\alpha\ \left(-l\ qHn+n\ qHn+\eta a-n\ \eta a\right)}{c\ (-1+n)+\alpha-l\ \alpha}$

Non strategic countries' eq'm efforts: $\frac{b\ (c\ (-1+n)+\alpha)+\alpha\ \left(\left(c-c\ n+(-1+l)\ \alpha\right)\ \eta-l\ \alpha\ \eta a\right)}{(c-\alpha)\ (c\ (-1+n)+\alpha)}$

Strategic country's true effort given non-strategic countries' effort with mistaken beliefs:

In[172]:=
$$\mathtt{FullSimplify}\Big[\mathtt{Solve}\Big[qLn == \frac{b\ (-1+n)+\alpha\ \left((n-l)\ \left(\frac{b\ (c\ (-1+n)+\alpha)+\alpha\ ((c-c\ n+(-1+l)\ \alpha)\ \eta-l\ \alpha\ \eta a)}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)+\eta-n\ \eta\right)}{c\ (-1+n)+\alpha-l\ \alpha},\ qLn\Big]\Big]$$

Out[172]=
$$\left\{\left\{qLn\to\frac{b\ (-1+n)+\alpha\ \left(\eta-n\ \eta+\frac{(-l+n)\ (b\ (c\ (-1+n)+\alpha)+\alpha\ ((c-c\ n+(-1+l)\ \alpha)\ \eta-l\ \alpha\ \eta a))}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)}{c\ (-1+n)+\alpha-l\ \alpha}\right\}\right\}$$

In[173]:=
$$\begin{aligned}&\mathtt{indutilnoncoopdeceta}[b\_,\ c\_,\ \alpha\_,\ \eta a\_,\ \eta\_,\ n\_,\ l\_] := \\ &\mathtt{FullSimplify}\Bigg[b\ \Bigg((n-l)\ \left(\frac{b\ (c\ (-1+n)+\alpha)+\alpha\ \left(\left(c-c\ n+(-1+l)\ \alpha\right)\ \eta-l\ \alpha\ \eta a\right)}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)+ \\ &\quad l\ \left(\frac{b\ (-1+n)+\alpha\ \left(\eta-n\ \eta+\frac{(-l+n)\ (b\ (c\ (-1+n)+\alpha)+\alpha\ (-(c\ (-1+n)+\alpha)\ \eta+l\ \alpha\ (-\eta a+\eta)))}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)}{c\ (-1+n)+\alpha-l\ \alpha}\right)\Bigg)- \\ &\quad (c/2)\ \left(\frac{b\ (-1+n)+\alpha\ \left(\eta-n\ \eta+\frac{(-l+n)\ (b\ (c\ (-1+n)+\alpha)+\alpha\ (-(c\ (-1+n)+\alpha)\ \eta+l\ \alpha\ (-\eta a+\eta)))}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)}{c\ (-1+n)+\alpha-l\ \alpha}\right)\text{^}2+ \\ &\quad \alpha\ \Bigg(\Bigg((n-l)\ \left(\frac{b\ (c\ (-1+n)+\alpha)+\alpha\ \left(\left(c-c\ n+(-1+l)\ \alpha\right)\ \eta-l\ \alpha\ \eta a\right)}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)+(l-1) \\ &\qquad \left(\frac{b\ (-1+n)+\alpha\ \left(\eta-n\ \eta+\frac{(-l+n)\ (b\ (c\ (-1+n)+\alpha)+\alpha\ ((c-c\ n+(-1+l)\ \alpha)\ \eta-l\ \alpha\ \eta a))}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)}{c\ (-1+n)+\alpha-l\ \alpha}\right)\Bigg)\Big/\ (n-1)-\eta\Bigg) \\ &\quad \left(\left(\frac{b\ (-1+n)+\alpha\ \left(\eta-n\ \eta+\frac{(-l+n)\ (b\ (c\ (-1+n)+\alpha)+\alpha\ (-(c\ (-1+n)+\alpha)\ \eta+l\ \alpha\ (-\eta a+\eta)))}{(c-\alpha)\ (c\ (-1+n)+\alpha)}\right)}{c\ (-1+n)+\alpha-l\ \alpha}\right)+1-\eta\right)\Bigg]\end{aligned}$$

Strategic country's optimal expectations:

In[174]:= `FullSimplify[Solve[D[indutilnoncoopdeceta[b, c, α, ηa, η, n, 1], ηa] == 0, ηa]]`

Out[174]=
$$\left\{\left\{\eta a\to\frac{c\ (c\ (-1+n)+\alpha)\ \left(b\ c\ (-1+n)-b\ (-2+n)\ \alpha+(c-\alpha)\ \alpha\right)+\alpha\ \left(-c^3\ (-1+n)-c^2\ \alpha+\alpha^3\right)\ \eta}{\alpha^4}\right\}\right\}$$

We evaluate these equations for a wide range of parameters and shared our findings in **`result 3 and the paragraph before.`**

In[175]:= `FullSimplify[Solve[D[indutilnoncoopdeceta[1, 3, 0.1, ηa, η, 5, 1], ηa] == 0, ηa]]`

Out[175]= $\left\{\left\{\eta a\to 4.35237\times 10^{6}-108\,899.\ \eta\right\}\right\}$

Heterogeneity: Self-Interested and Reciprocal Countries

There are A<N reciprocal countries with α_i = α and N-A self-interested countries with α_i = 0.

Non-Cooperative Game - Proposition 5

Self-interested countries have a dominant strategy to exert b/c. Let's find out about the reciprocal Nash efforts. To do that, let's write the utility function.

In[176]:= ui[b_, c_, n_, a_, α_, η_, qi_, qj_] :=

$$b\left((n-a)\,\frac{b}{c}+(a-1)\,qj+qi\right)-\frac{c}{2}\,qi^2+\alpha\left(\frac{(n-a)\,\frac{b}{c}+(a-1)\,qj}{n-1}-\eta\right)(qi+1-\eta)$$

Taking derivative wrt qi and solving for qi yields the Nash efforts of the reciprocal countries:

In[177]:= D[ui[b, c, n, a, α, η, qi, qj], qi]

Out[177]= $$b-c\,qi+\alpha\left(\frac{\frac{b\,(-a+n)}{c}+(-1+a)\,qj}{-1+n}-\eta\right)$$

In[178]:= $$\text{FullSimplify}\left[\text{Solve}\left[b-c\,qi+\alpha\left(\frac{\frac{b\,(-a+n)}{c}+(-1+a)\,qi}{-1+n}-\eta\right)==0,\ qi\right]\right]$$

Out[178]= $$\left\{\left\{qi\to\frac{b\,c\,(-1+n)+b\,(-a+n)\,\alpha-c\,(-1+n)\,\alpha\,\eta}{c\,(c\,(-1+n)+\alpha-a\,\alpha)}\right\}\right\}$$

In[179]:= $$\text{qnc}[b\_,c\_,n\_,a\_,\alpha\_,\eta\_]:=\frac{b\,c\,(-1+n)+b\,(-a+n)\,\alpha-c\,(-1+n)\,\alpha\,\eta}{c\,(c\,(-1+n)+\alpha-a\,\alpha)}$$

What is the impact of α on qnc? Similar to the homogenous case, whether η > b/c holds determines the sign of the impact.

In[180]:= FullSimplify[D[qnc[b, c, n, a, α, η], α]]

Out[180]= $$\frac{(-1+n)^2\,(b-c\,\eta)}{(c\,(-1+n)+\alpha-a\,\alpha)^2}$$

The level of reciprocal concern α needed for the reciprocal countries to exert full effort:

In[183]:= FullSimplify[Solve[qnc[b, c, n, a, α, η] == 1, α]]

Out[183]= $$\left\{\left\{\alpha\to\frac{(b-c)\,c\,(-1+n)}{a\,(b-c)+c-b\,n+c\,(-1+n)\,\eta}\right\}\right\}$$

The level of reciprocal concern α needed for the reciprocal countries to exert zero effort:

In[184]:= FullSimplify[Solve[qnc[b, c, n, a, α, η] == 0, α]]

Out[184]= $$\left\{\left\{\alpha\to\frac{b\,c\,(-1+n)}{a\,b-b\,n+c\,(-1+n)\,\eta}\right\}\right\}$$

Partial Cooperation - Result 4

Reciprocal Non-signatories

Remember that self-interested non-signatories have dominant strategy of exerting b/c.
kr and ks refer to the number of reciprocal and self-interested signatories.

In[189]:=
```
urnsi[b_, c_, n_, a_, ks_, kr_, α_, η_, qi_, qss_, qrs_, qrns_] :=
```
$$b\left((n-a-ks)\frac{b}{c}+ks*qss+kr*qrs+(a-kr-1)\,qrns+qi\right)-\frac{c}{2}qi^2+\alpha\left(\frac{(n-a-ks)\frac{b}{c}+ks*qss+kr*qrs+(a-kr-1)\,qrns}{n-1}-\eta\right)(qi+1-\eta)$$

Taking derivative wrt qi yields:

In[190]:=
```
D[urnsi[b, c, n, a, ks, kr, α, η, qi, qss, qrs, qrns], qi]
```

Out[190]=
$$b-c\,qi+\alpha\left(\frac{\frac{b(-a-ks+n)}{c}+(-1+a-kr)\,qrns+kr\,qrs+ks\,qss}{-1+n}-\eta\right)$$

Using symmetry and solving for qrns yields the result:

In[191]:=
```
FullSimplify[
```
$$\text{Solve}\left[b-c\,qrns+\alpha\left(\frac{\frac{b(-a-ks+n)}{c}+(a-kr-1)\,qrns+kr\,qrs+ks\,qss}{-1+n}-\eta\right)==qrns,\ qrns\right]\right]$$

Out[191]=
$$\left\{\left\{qrns\to\frac{b\,c\,(-1+n)-b\,(a+ks-n)\,\alpha+c\,\alpha\,(kr\,qrs+ks\,qss+\eta-n\,\eta)}{c\,(-1+c\,(-1+n)+n+\alpha-a\,\alpha+kr\,\alpha)}\right\}\right\}$$

In[192]:=
```
qrns[b_, c_, n_, a_, ks_, kr_, α_, η_, qss_, qrs_] :=
```
$$\frac{b\,c\,(-1+n)-b\,(a+ks-n)\,\alpha+c\,\alpha\,(kr\,qrs+ks\,qss+\eta-n\,\eta)}{c\,(-1+c\,(-1+n)+n+\alpha-a\,\alpha+kr\,\alpha)}$$

Signatories

Let’s start with the self-interested signatories problem.

In[193]:=
```
usig[b_, c_, n_, a_, ks_, kr_, α_, η_, qss_, qrs_] :=
  FullSimplify[ks (b ((n - a - ks) b/c + (a - kr) qrns[b, c, n, a, ks, kr, α, η, qss, qrs] +
        ks * qss + kr * qrs) - c/2 qss^2) +
    kr (b ((n - a - ks) b/c + (a - kr) qrns[b, c, n, a, ks, kr, α, η, qss, qrs] +
        ks * qss + kr * qrs) - c/2 qrs^2 -
      + α (1/(n - 1) ((n - a - ks) b/c + (a - kr) qrns[b, c, n, a, ks, kr, α, η, qss, qrs] +
          ks * qss + (kr - 1) qrs) - η) (qrs + 1 - η))]
```

Take first derivative wrt qss (FOC) and solving for qss (self-interested signatory effort); and then similarly with qrs (reciprocal signatory effort).

In[194]:=
```
FullSimplify[Solve[D[usig[b, c, n, a, ks, kr, α, η, qss, qrs], qss] == 0, qss]]
```

Out[194]=

$$\left\{\left\{\text{qss} \to \frac{((1+c)\,(-1+n)+\alpha)\,\left(b\,(\text{kr}+\text{ks})\,(-1+n)+\text{kr}\,\alpha\,(-1-\text{qrs}+\eta)\right)}{c\,(-1+n)\,\left(-1+c\,(-1+n)+n+\alpha-a\,\alpha+\text{kr}\,\alpha\right)}\right\}\right\}$$

In[195]:=
```
FullSimplify[Solve[D[usig[b, c, n, a, ks, kr, α, η, qss, qrs], qrs] == 0, qrs]]
```

Out[195]=

$$\Big\{\Big\{\text{qrs} \to \Big(b\,\big(c\,(1+c)\,(\text{kr}+\text{ks})\,(-1+n)^2+(-1+n)\,\big(a+\text{ks}+2\,c\,(\text{kr}+\text{ks})-n-c\,n\big)\,\alpha+(a+\text{ks}-n)\,\alpha^2\big)+ c\,\alpha\,\big(-1+n+\text{ks}\,\text{qss}+\alpha-a\,\alpha-\text{ks}\,\text{qss}\,(n+\alpha)+\text{kr}\,(-1+n)\,(-1+\eta)+2\,\eta+ (-3+n)\,n\,\eta+(-2+a+n)\,\alpha\,\eta+c\,(-1+n)\,\big(1-\text{kr}-\text{ks}\,\text{qss}+(-2+\text{kr}+n)\,\eta\big)\big)\Big)\Big/ \Big(c\,\big(c\,(1+c)\,(-1+n)^2-\big(2+c\,(1+a-3\,\text{kr})-2\,\text{kr}\big)\,(-1+n)\,\alpha+2\,(-1+a)\,\alpha^2\big)\Big)\Big\}\Big\}$$

Substituting qrs into qss and solving for qss yields

```
In[196]:= FullSimplify[Solve[
    (((1 + c) (-1 + n) + α) (b (kr + ks) (-1 + n) + kr α (-1 - ((b (c (1 + c) (kr + ks) (-1 + n)^2 +
                  (-1 + n) (a + ks + 2 c (kr + ks) - n - c n) α + (a + ks - n) α^2) + c α
                (-1 + n + ks qss + α - a α - ks qss (n + α) + kr (-1 + n) (-1 + η) + 2 η + (-3 + n)
                   n η + (-2 + a + n) α η + c (-1 + n) (1 - kr - ks qss + (-2 + kr + n) η))) /
             (c (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α +
                  2 (-1 + a) α^2))) + η))) /
      (c (-1 + n) (-1 + c (-1 + n) + n + α - a α + kr α)) == qss, qss]]
```

```
Out[196]= {{qss →
     ((1 - n) ((1 + c) (-1 + n) + α) (b (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) + c^2
               (kr + ks) (-1 + n)^2 (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α
               ((-2 + kr) (kr + ks) (-1 + n) + 2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α
            (c^2 (-1 + n)^2 (-1 + η) - c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n)
                  α η) + α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
      (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
          kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (-(1 + c) (-1 + n) + (-1 + a - kr) α)
            (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2)))}}
```

Self-interested signatory effort

```
In[197]:= qss[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
   ((1 - n) ((1 + c) (-1 + n) + α) (b (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) +
            c^2 (kr + ks) (-1 + n)^2 (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α
              ((-2 + kr) (kr + ks) (-1 + n) + 2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α
          (c^2 (-1 + n)^2 (-1 + η) - c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n) α η) +
            α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
    (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
        kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (-(1 + c) (-1 + n) + (-1 + a - kr) α)
          (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2)))
```

Substitute qss into qrs

In[199]:=
```
FullSimplify[
 (b (c (1 + c) (kr + ks) (-1 + n)^2 + (-1 + n) (a + ks + 2 c (kr + ks) - n - c n) α + (a + ks - n) α^2) +
    c α (-1 + n + ks (qss[b, c, n, a, ks, kr, α, η]) + α - a α - ks (qss[b, c, n, a, ks, kr,
            α, η]) (n + α) + kr (-1 + n) (-1 + η) + 2 η + (-3 + n) n η + (-2 + a + n) α η +
         c (-1 + n) (1 - kr - ks (qss[b, c, n, a, ks, kr, α, η]) + (-2 + kr + n) η))) /
  (c (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2))]
```

Out[•]=
```
(b (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
            (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
          (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
              (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
          (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
   α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) +
        c^2 (-1 + n)^2 (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α +
            2 (-1 + n) (-2 + kr + n) η - (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) +
        c (-1 + n) (kr^2 (-1 + n) α (-1 + η) + (-1 + n + α - a α)
              (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
            kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
                α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
 (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3
     α +
    (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
     (-1 + n)^2 α^2 -
    2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 -
    kr ks α^4)
```

Reciprocal signatory effort

In[200]:=
```
qrs[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
  (b (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
            (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
          (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
              (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
          (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
     α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) + c^2 (-1 + n)^2
            (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α + 2 (-1 + n) (-2 + kr + n) η -
              (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) + c (-1 + n) (kr^2 (-1 + n) α
               (-1 + η) + (-1 + n + α - a α) (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
              kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
                  α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
   (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3 α +
      (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
       (-1 + n)^2 α^2 -
      2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 - kr ks α^4)
```

We can substitute signatory efforts into qrns (Reciprocal non-signatory effort)

In[201]:=
```
qrnsf[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
 FullSimplify[qrns[b, c, n, a, ks, kr, α, η, ((1 - n) ((1 + c) (-1 + n) + α)
      (b (c^3 (kr + ks) (-1 + n)^3 - kr (a + ks - n) α^2 (-1 + n + α) + c^2 (kr + ks) (-1 + n)^2
            (-1 + n - (1 + a - 2 kr) α) + c (-1 + n) α ((-2 + kr) (kr + ks) (-1 + n) +
              2 (-1 + a - kr) (kr + ks) α + kr n α)) + c kr α (c^2 (-1 + n)^2 (-1 + η) -
          c (-1 + n) (-1 + n - a α + 2 kr α + η - n η + (-1 + a - 2 kr + n) α η) +
          α (-1 + n + α - a α + kr (-1 + n) (-1 + η) + a α η - n (-1 + n + α) η)))) /
     (c (-1 + n) (-kr ks α^2 ((1 + c) (-1 + n) + α) + c kr ks (-1 + n) α^2 ((1 + c) (-1 + n) + α) +
         kr ks α^2 ((1 + c) (-1 + n) + α) (n + α) - c (1 - n) (- (1 + c) (-1 + n) + (-1 + a - kr) α)
          (c (1 + c) (-1 + n)^2 - (2 + c (1 + a - 3 kr) - 2 kr) (-1 + n) α + 2 (-1 + a) α^2))),
    (b (-1 + n) (c (1 + c)^2 (kr + ks) (-1 + n)^3 - (1 + c) (-1 + n)^2
          (-ks + (kr + ks) (ks + c (-3 - kr + ks)) + a (-1 + c (kr + ks)) + n + c n) α -
         (-1 + n) (a^2 + ks (-2 + kr + 2 ks) + a (-2 - kr + ks + c (-1 + 2 kr + 2 ks - n) - n) +
            (2 + kr) n + c (ks (-3 + 2 ks) + 2 n + kr (-2 - 2 kr + n))) α^2 -
         (a^2 - ks + ks^2 + n + kr n - a (1 + kr - ks + n)) α^3) +
      α (-kr ks α (-1 + n + α)^2 (-1 + η) + c^3 (-1 + n)^3 (1 - kr + (-2 + kr + n) η) +
         c^2 (-1 + n)^2 (-2 + 2 kr + 2 n - 2 kr n + 2 α - 2 a α + a kr α - kr^2 α + kr ks α +
            2 (-1 + n) (-2 + kr + n) η - (4 + kr - 2 n + a (-3 + kr + n) - kr (kr - ks + n)) α η) +
         c (-1 + n) (kr^2 (-1 + n) α (-1 + η) + (-1 + n + α - a α)
             (-1 + n + α - a α + (2 + (-2 + a) α + n (-3 + n + α)) η) +
            kr (-1 + η + n^2 (-1 + η + α η) + n (2 + a α + 2 ks α + (-2 - (2 + a + 2 ks) α + α^2) η) +
               α (α + a (1 + α) (-1 + η) + η - 2 α η + 2 ks (-1 + α + η - α η)))))) /
     (c^2 (1 + c)^2 (-1 + n)^4 - 2 c (1 + c) (1 + a c - (1 + 2 c) kr) (-1 + n)^3 α +
      (c ((-1 + a) (4 + (3 + a) c) - 2 (a - c + 2 a c) kr + (2 + 3 c) kr^2) - (1 + c)^2 kr ks)
       (-1 + n)^2 α^2 - 2 ((-1 + a) c (-1 + a - kr) + (1 + c) kr ks) (-1 + n) α^3 - kr ks α^4)]]
```

Indirect Utility Functions

Under symmetry, we had two stability conditions and two indirect utility functions. Under asymmetry, we have 4 of them.

Self-interested non-signatory

In[212]:=
```
indusns[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
 FullSimplify[b ((n - a - ks) b/c + (a - kr) qrnsf[b, c, n, a, ks, kr, α, η] +
      ks * qss[b, c, n, a, ks, kr, α, η] + kr * qrs[b, c, n, a, ks, kr, α, η]) - c/2 (b/c)^2]
```

Reciprocal non-signatory

In[202]:=
```
indurns[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
  b ((n - a - ks) b/c + (a - kr) qrnsf[b, c, n, a, ks, kr, α, η] +
      ks * qss[b, c, n, a, ks, kr, α, η] + kr * qrs[b, c, n, a, ks, kr, α, η]) -
   c/2 qrnsf[b, c, n, a, ks, kr, α, η]^2 +
   α (1/(n - 1) ((n - a - ks) b/c + (a - kr - 1) qrnsf[b, c, n, a, ks, kr, α, η] +
         ks * qss[b, c, n, a, ks, kr, α, η] + kr * qrs[b, c, n, a, ks, kr, α, η]) - η)
    (qrnsf[b, c, n, a, ks, kr, α, η] + 1 - η)
```

Self-interested signatory

In[203]:=
```
induss[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
  b ((n - a - ks) b/c + (a - kr) qrnsf[b, c, n, a, ks, kr, α, η] +
      ks * qss[b, c, n, a, ks, kr, α, η] + kr * qrs[b, c, n, a, ks, kr, α, η]) -
   c/2 (qss[b, c, n, a, ks, kr, α, η])^2
```

Reciprocal signatory

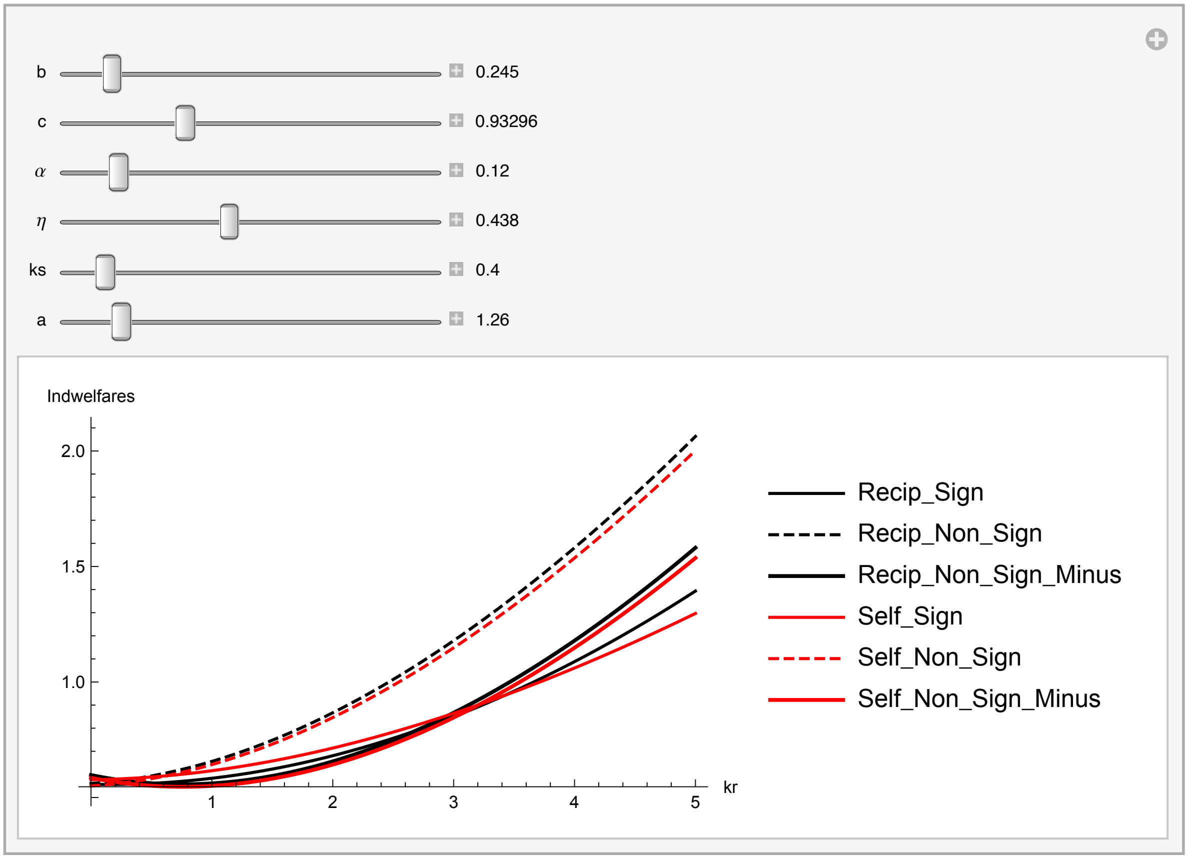
In[204]:=
```
indurs[b_, c_, n_, a_, ks_, kr_, α_, η_] :=
  b ((n - a - ks) b/c + (a - kr) qrnsf[b, c, n, a, ks, kr, α, η] +
      ks * qss[b, c, n, a, ks, kr, α, η] + kr * qrs[b, c, n, a, ks, kr, α, η]) -
   c/2 (qrs[b, c, n, a, ks, kr, α, η])^2 + α (1/(n - 1) ((n - a - ks) b/c +
        (a - kr) qrnsf[b, c, n, a, ks, kr, α, η] + ks * qss[b, c, n, a, ks, kr, α, η] +
        (kr - 1) * qrs[b, c, n, a, ks, kr, α, η]) - η) (qrs[b, c, n, a, ks, kr, α, η] + 1 - η)
```

Stability Conditions (Result 4)

We employ the following figures that we manipulate and numerically analyze the stability conditions for a wide range of parameters. We summarized our results in the paper.

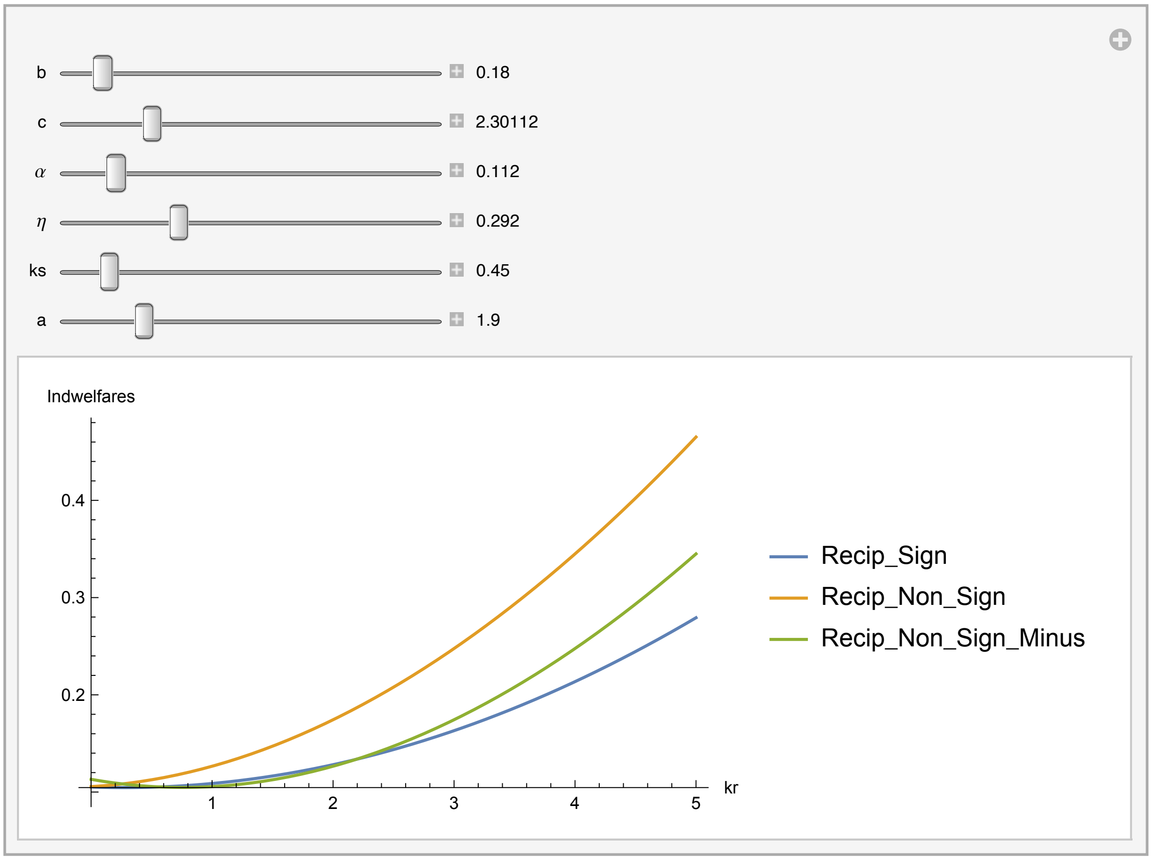
In[205]:=
```
Manipulate[Plot[{indurs[b, c, 10, a, ks, kr, α, η], indurns[b, c, 10, a, ks, kr, α, η],
   indurns[b, c, 10, a, ks, kr - 1, α, η], induss[b, c, 10, a, ks, kr, α, η],
   indusns[b, c, 10, a, ks, kr, α, η], indusns[b, c, 10, a, ks, kr - 1, α, η]},
  {kr, 0, 5}, AxesLabel → {kr, "Indwelfares"}, PlotLegends →
   {"Recip_Sign", "Recip_Non_Sign", "Recip_Non_Sign_Minus", "Self_Sign",
    "Self_Non_Sign", "Self_Non_Sign_Minus"}, PlotRange → Full, PlotStyle →
   {Black, {Black, Dashed}, {Black, Thick}, Red, {Red, Dashed}, {Red, Thick}} ],
 {b, 0, 2.5, Appearance → "Labeled"}, {c, b, 10 b, Appearance → "Labeled"},
 {α, 0, 1, Appearance → "Labeled" },  {η, 0, 1, Appearance → "Labeled"},
 {ks, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[205]=



In[206]:=
```
Manipulate[Plot[{indurs[b, c, 10, a, ks, kr, α, η],
    indurns[b, c, 10, a, ks, kr, α, η], indurns[b, c, 10, a, ks, kr - 1, α, η]},
   {kr, 0, 5}, AxesLabel → {kr, "Indwelfares"}, PlotLegends →
    {"Recip_Sign", "Recip_Non_Sign", "Recip_Non_Sign_Minus"}, PlotRange → Full ],
 {b, 0, 2.5, Appearance → "Labeled"}, {c, b, 10, Appearance → "Labeled"},
 {α, 0, 1, Appearance → "Labeled" },  {η, 0, 1, Appearance → "Labeled"},
 {ks, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[206]=

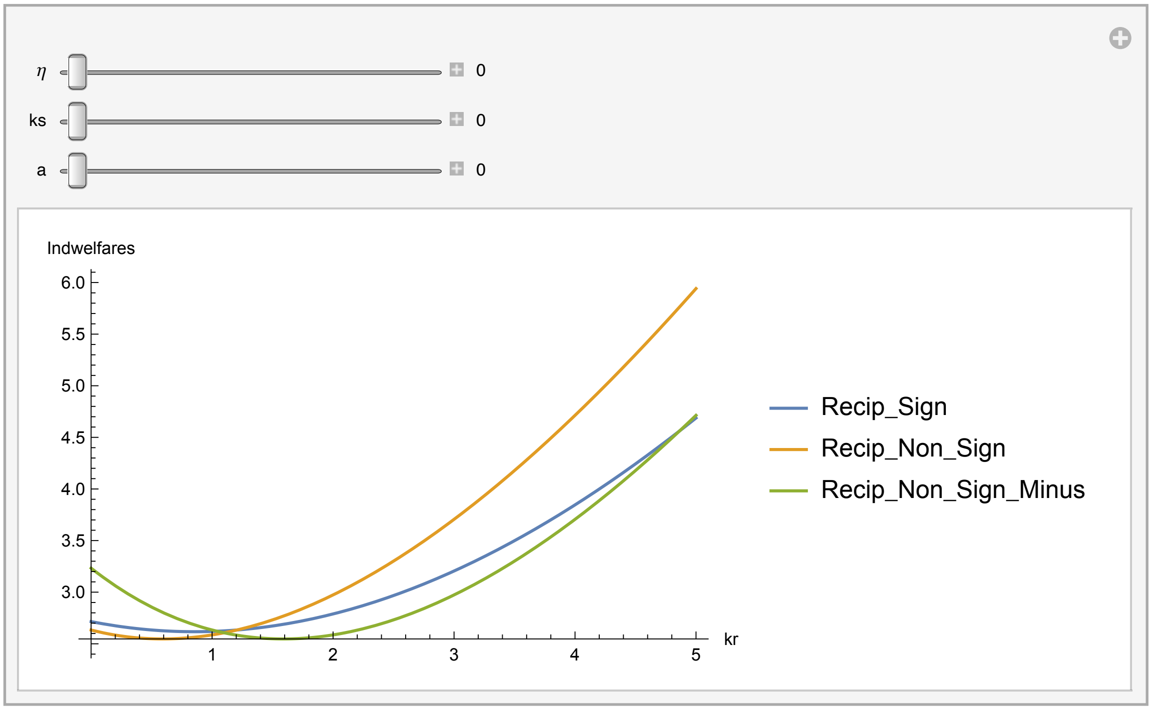


In[207]:=
```
Manipulate[Plot[{induss[b, c, 10, a, ks, kr, α, η],
    indusns[b, c, 10, a, ks, kr, α, η], indusns[b, c, 10, a, ks - 1, kr, α, η]},
   {ks, 0, 5}, AxesLabel → {kr, "Indwelfares"}, PlotLegends →
    {"Self_Sign", "Self_Non_Sign", "Self_Non_Sign_Minus"}, PlotRange → Full ],
  {b, 0, 2.5, Appearance → "Labeled"}, {c, b, 10, Appearance → "Labeled"},
  {α, 0, 1, Appearance → "Labeled" },  {η, 0, 1, Appearance → "Labeled"},
  {kr, 0, 5, Appearance → "Labeled"}, {a, 0, 10, Appearance → "Labeled"}]
```

Out[207]=

In[208]:=
```
Manipulate[Plot[{indurs[1, 4, 10, a, ks, kr, 0.8, η],
    indurns[1, 4, 10, a, ks, kr, 0.8, η], indurns[1, 4, 10, a, ks, kr - 1, 0.8, η]},
   {kr, 0, 5}, AxesLabel → {kr, "Indwelfares"}, PlotLegends →
    {"Recip_Sign", "Recip_Non_Sign", "Recip_Non_Sign_Minus"}, PlotRange → Full],
  {η, 0, 1, Appearance → "Labeled"}, {ks, 0, 5, Appearance → "Labeled"},
  {a, 0, 10, Appearance → "Labeled"}]
```

Out[208]=



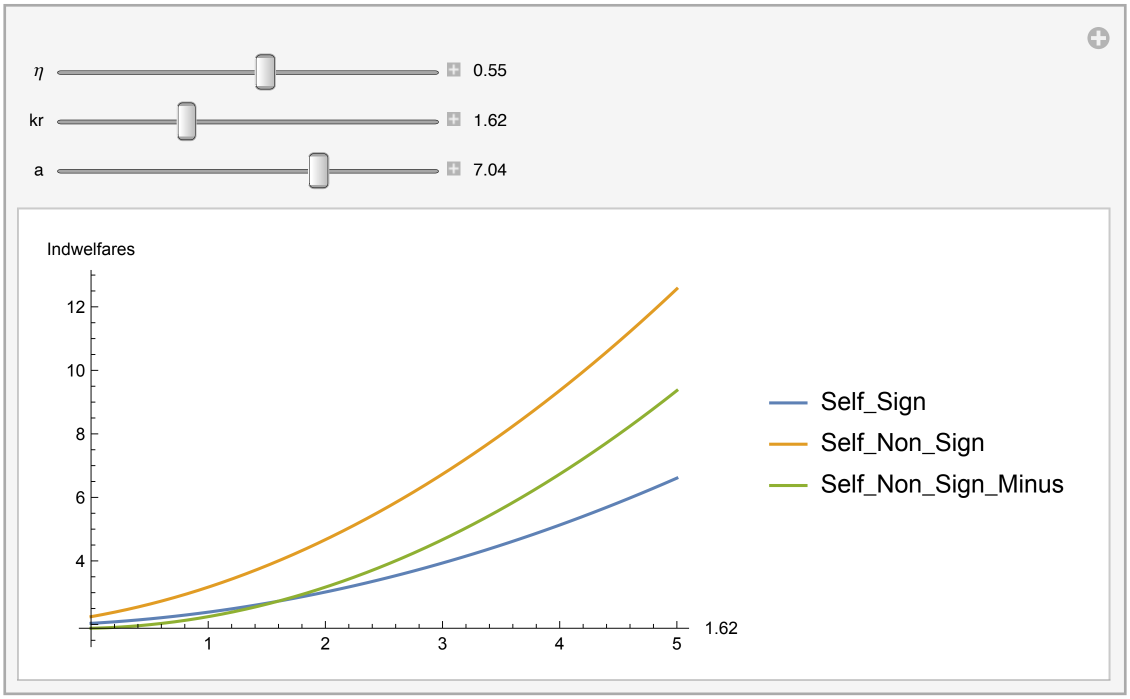
In[213]:=
```
Solve[
 indurs[1, 4, 10, 5, 0, kr, 0.8, 0.1] == indurns[1, 4, 10, 5, 0, kr - 1, 0.8, 0.1], kr]
```

Solve: Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[213]=
```
{{kr → -52.3775}, {kr → -13.4121 - 0.284697 i},
 {kr → -13.4121 + 0.284697 i}, {kr → 0.986896}, {kr → 4.44768}, {kr → 141.873}}
```

In[210]:=
```
Manipulate[Plot[{induss[1, 4, 10, a, ks, kr, 0.8, η],
    indusns[1, 4, 10, a, ks, kr, 0.8, η], indusns[1, 4, 10, a, ks - 1, kr, 0.8, η]},
   {ks, 0, 5}, AxesLabel → {kr, "Indwelfares"}, PlotLegends →
    {"Self_Sign", "Self_Non_Sign", "Self_Non_Sign_Minus"}, PlotRange → Full],
  {η, 0, 1, Appearance → "Labeled"}, {kr, 0, 5, Appearance → "Labeled"},
  {a, 0, 10, Appearance → "Labeled"}]
```

Out[210]=



In[214]:=
```
Solve[
  induss[1, 4, 10, 5, ks, 0, 0.5, 0.5] == indusns[1, 4, 10, 5, ks - 1, 0, 0.5, 0.5], ks]
```

Out[214]= {{ks → 0.998492}, {ks → 3.00151}}

In[215]:= `qss[1, 4, 10, 5, 3, 0, 0.1, 0.5]`

Out[215]= 0.758408

In[216]:= `qrs[1, 4, 10, 5, 3, 0, 0.1, 0.5]`

Out[216]= 0.766095

1. eta = 0.5 and a = 5. Let's start from the case where kr=ks=0 and check each type of country's incentives to become a signatory.
ks = 0 -> kr = 3 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 - > ks = 3 -> kr =0
Stable coalition: (ks*, kr*) = (3,0)

2. eta = 0.2 and a = 5.
ks= 0 -> kr = 3 -> ks =0.
ks = 1 -> kr =2 -> ks = 1
ks = 2 -> kr = 1 -> ks = 2
ks = 3 -> kr = 0 -> ks = 3
Stable Coalitions: (ks*, kr*) = {(0,3),(1,2),(2,1), (3,0)}

```
3. eta = 0.8 and a = 5.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 2
ks = 2 -> kr = 0 -> ks = 3 -> kr = 0
Stable Coalition: (ks*, kr*) = {(1, 2),(3, 0)}

------
4. eta = 0.5 and a = 8.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3, but
there are no 3 self-interested.
No Stable Coalition

5. eta = 0.2 and a = 8.
ks = 0 -> kr = 3 -> ks = 0
ks = 1 -> kr = 2 -> ks = 1
ks = 2 -> kr = 1 -> ks = 2
Stable Coalitions: (ks*, kr*) = {(0,3),(1,2),(2,1)}

6. eta = 0.8 and a = 8.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3, but
there are no 3 self-interested.
No Stable Coalition
-----
7. eta = 0.5 and a = 2.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3, kr = 0.
Stable Coalition: (ks*, kr*) = (3,0)

8. eta = 0.2 and a = 2.
ks = 0 -> kr = 3, but there are no 3 reciprocal.
ks = 1 -> kr = 2 -> ks = 1
ks = 2 -> kr = 1 -> ks = 2
ks = 3 -> kr = 1 -> ks = 3
Stable Coalitions: (ks*, kr*) = {(1,2),(2,1),(3,0)}

9. eta = 0.8 and a = 2.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 -> ks = 3 -> kr = 0
Stable Coalitions: (ks*, kr*) = {(3,0)}

Let’s also analyze 1-3 under the assumption that alpha = 0.5.
1a. eta = 0.5 and a = 5.
ks = 0 -> kr = 2 -> ks = 1 -> kr = 1 -> ks = 2 -> kr = 0 - > ks = 3 -> kr =0
Stable coalition: (ks*, kr*) = (3,0)

2a. eta = 0.2 and a = 5.
ks= 0 -> kr = 3 -> ks  =0.
ks = 1 -> kr =2 -> ks = 1
ks = 2 -> kr = 1 -> ks = 2
ks = 3 -> kr = 0 -> ks = 3

Stable Coalitions: (ks*, kr*) = {(0,3),(1,2),(2,1), (3,0)}

3a. eta = 0.8 and a = 5.
```

```
ks = 0 -> kr = 2 -> ks = 1 -> kr = 2
ks = 2 -> kr = 0 -> ks = 3 -> kr = 0
Stable Coalition: (ks*, kr*) = {(1, 2),(3, 0)}
```